

$$\begin{array}{c}
 f: M \rightarrow M \rightarrow \text{surjection} \\
 \uparrow \quad \quad \quad (\text{injection}) \\
 \text{Noetherian (Artin)} \rightarrow f: \text{iso}
 \end{array}$$

Artinian ring

$$d_{cc}$$

$$\mathbb{Z} \supset (2) \supset (2^2) \supset (2^3) \supset \dots$$

$$M / \ker f \cong M$$

$$\ker T^n \cap \text{Im } T^n$$

$$\ker T^n$$

$$0 \xrightarrow{f} L \xrightarrow{f} M \xrightarrow{g} N \rightarrow 0$$

$$0 \rightarrow L \xrightarrow{f} M \hookrightarrow \text{f: inj}$$

$$M \xrightarrow{g} N \rightarrow 0 \Leftrightarrow \text{f: sur}$$

$$\begin{array}{ccc}
 M : \text{f.f.} & \Leftrightarrow & L \hookrightarrow N : \text{f.f.} \\
 (\text{Artin}) & & (\text{Artin})
 \end{array}$$

$$M / f(L) \cong N$$

$$\bigoplus L \cong N$$

$$\text{(Cor)} M_i : \text{f.f.} \quad (1 \leq i \leq n)$$

$$\Rightarrow \bigoplus_{i=1}^n M_i : \text{f.f.}$$

$$0 \rightarrow M_1 \rightarrow \bigoplus_{i=1}^n M_i \rightarrow \bigoplus_{i=1}^{n-1} M_i \rightarrow 0$$

$T: V \rightarrow V$ is diagonalizable $\Leftrightarrow m_T$ splits and separable

" triangulizable $\Leftrightarrow m_T$ is separable

E/F is separable $\Leftrightarrow \# \text{Emb}_F(E, \bar{F}) = [E:F]$

E/F is splitting $\Leftrightarrow \text{Emb}_F(E, \bar{F}) = \text{Aut}(E/F)$

relation $\{ \}$

$$\alpha, \beta \in \mathbb{R}^3, \quad r \in \mathbb{R}^3,$$

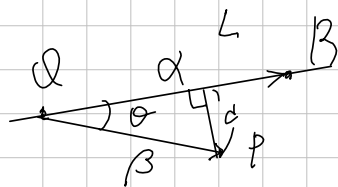
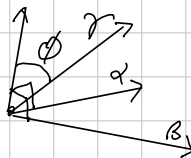
$$1. |\alpha \times \beta| = |\alpha| \cdot |\beta| \cdot \sin \theta.$$

$$2. \alpha \cdot \beta = |\alpha| \cdot |\beta| \cdot \cos \theta.$$

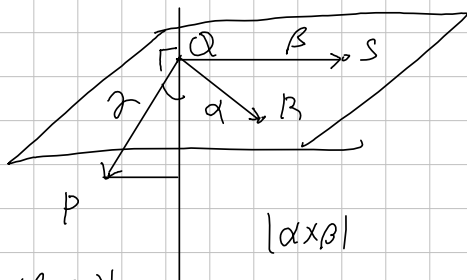
$$3. V = A \cdot h = \underbrace{|\alpha| \cdot |\beta| \cdot \sin \theta}_{|\alpha \times \beta|} \cdot \underbrace{h}_{|r| \cdot \cos \phi}$$

$$= |\alpha \times \beta| \cdot |r| \cdot \cos \phi$$

$$= |r \cdot (\alpha \times \beta)|.$$



$$d = \frac{|\alpha \times \beta|}{|\alpha|}.$$



$$d = \frac{|\alpha \cdot (\beta \times r)|}{|\alpha \times \beta|}$$

$$L: \mathbb{R} \rightarrow \mathbb{R}^3: t \mapsto (a \ b \ c) \cdot (t) = \begin{pmatrix} at \\ bt \\ ct \end{pmatrix}$$

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}: (x, y, z) \mapsto (a \ b \ c) \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$n = (a, b, c), \text{ and } P_0 = (x_0, y_0, z_0).$$

$$n \cdot (P_0 - P) = 0 \Leftrightarrow a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

1). $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist.

Proof. Take $f(x,y) = \frac{x^2 - y^2}{x^2 + y^2}$.

First, observe that $\lim_{y \rightarrow 0} f(0,y) = -1$

but, $\lim_{x \rightarrow 0} f(x,0) = 1$. So that proved.

2). $f(x,y) = \frac{xy}{x^2 + y^2}$

Observe that $\lim_{x \rightarrow 0} f(x,x) = \frac{1}{2}$

but $\lim_{x \rightarrow 0} f(x,0) = 0$ $a+b \geq 2\sqrt{ab}$

3). $f(x,y) = \frac{x^2 y}{x^2 + y^2}$

Given $\epsilon > 0$, take $\delta = \epsilon$. Then, $\|(x,y)\| < \delta \Rightarrow$

$$|f(x,y) - 0| = \left| \frac{x^2 y}{x^2 + y^2} \right| \leq \frac{|x^2 y|}{2|x y|} = \frac{1}{2} |x| < \delta = \epsilon.$$